Real Variables II - MATH6321

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Contents

Contents **1**

1 Hilbert Spaces (continuation) **3**

1.1 Orthonormal Sets 3

2 Banach Space Techniques **10**

Course Information

14/01/2025 Office : Tuesday 11:30 - 12:30 PM, Wednesday 1-2PM
Midterm: 4th March 2025, in class

Chapter 1

Hilbert Spaces (continuation)

Exercise 1.0.1 (Warm up)**.** Show that if $f \in \ell^2(X)$, then utmost countably many values of f are non-zero.

Solution 1.0.1. *Take the sets*

$$E_n = \{x \in X \ : \ |f(x)| > 1/n\}$$

and consider their union.

1.1 Orthonormal Sets

Definition 1.1.1. A orthonormal family in a Hilbert space is a family $(u_\alpha)_{\alpha \in A}$ such that $\langle u_\alpha, u_\beta \rangle = 0$ if $\alpha \neq \beta$ and $\|u_\alpha\| = 1$ for each $\alpha, \beta \in A$.

Theorem 1.1.1 (Bessel's Inequality)**.** *Given an orthonormal family $(u_\alpha)_{\alpha \in A}$ in a Hilbert space $\mathcal{H}$, and $h \in \mathcal{H}$, then*

$$\|h\|^2 \geq \sum_{\alpha \in A} |\langle h, u_\alpha \rangle|^2$$

Proof. Let $B \subset A$ be finite, and let

$$g = \sum_{\alpha \in B} \langle h, u_\alpha \rangle u_\alpha$$

We can easily show that, $\langle h - g, g \rangle = 0$. Thus

$$\begin{aligned} \|h\|^2 &= \|h - g + g\|^2 \\ &= \langle h - g + g, h - g + g \rangle \\ &= \|h - g\|^2 + \|g\|^2 \\ &\geq \|g\|^2 \end{aligned}$$

Now the inequality follows form the definition of summation as the supremum of finite index sums. $\square$

Definition 1.1.2. Let $\mathcal{H}$ be a Hilbert space. An orthonormal family $(u_\alpha)_{\alpha \in A}$ is called complete, or an orthonormal basis, if for each $h \in H$,

$$\|h\|^2 = \sum_{\alpha \in A} |\langle h, u_\alpha \rangle|^2$$

Definition 1.1.3. A set $U = \{u_\alpha \ : \ \alpha \in A\}$ is a maximal orthonormal set if for any V with $V \supset U$ and V is orthonormal, then $V = U$.

Theorem 1.1.2. *Let $\mathcal{H}$ be a Hilbert space, $(u_\alpha)_{\alpha \in A}$ an orthonormal family, then the following are equivalent.*

1. *$(u_\alpha)_{\alpha \in A}$ is an orthonormal basis*
2. *span$\{u_\alpha \ : \ \alpha \in A\}$ is dense in $\mathcal{H}$*
3. *$\{u_\alpha\}$ is a maximal orthonormal set*

Proof. ($1 \implies 2$) Let h be given. Consider for any $B \subset A$,

$$g = \sum_{\alpha \in B} \langle h, u_\alpha \rangle u_\alpha$$

then we recall $\langle h - g, g \rangle = 0$. And thus

$$\|h\|^2 = \|h - g\|^2 + \|g\|^2$$

We know from equality in Bessel's inequality that, for given $\varepsilon > 0$ we can choose B such that $\|h\|^2 - \|g\|^2 < \varepsilon^2$. Hence

$$\|h - g\|^2 = \|h\|^2 - \|g\|^2 < \varepsilon^2$$

Thus, we can find a g that is arbitrarily close to h.

($\neg 3 \implies \neg 2$). Assuming 3 is wrong, there exists $u \in \mathcal{H}$, such that $\|u\| = 1$, and $\langle u, u_\alpha \rangle = 0$ for each $\alpha \in A$. Next, for any finite $B \subset A$, and any $c_\alpha \in \mathbb{C}$, we look at

$$\begin{aligned} \|u - \sum_{\alpha \in B} c_\alpha u_\alpha\| &= \|u\|^2 + \|\sum_{\alpha \in B} c_\alpha u_\alpha\|^2 \\ &= \|u\|^2 + \sum_{\alpha \in B} |c_\alpha|^2 \\ &\geq \|u\|^2 = 1 \end{aligned}$$

Thus $u \notin \overline{\text{span}\{u_\alpha\}}$.

($\neg 1 \implies \neg 3$). Assume there is $h \in \mathcal{H}$ such that

$$\|h\|^2 > \sum_{\alpha \in A} |\langle h, u_\alpha \rangle|^2$$

We know that $A_o = \{\alpha \in A \ : \ \langle h, u_\alpha \rangle \neq 0\}$ is at most countable from Exercise 1.0.1. We can find $A_1 \subset A_2 \subset \ldots$ each finite where

$$A_n = \{\alpha \in A_o \ : \ |\langle h, u_\alpha \rangle| \geq \frac{1}{n}\}$$

and

$$A_o = \bigcup_{n=1}^{\infty} A_n$$

Let $g_n = \sum_{\alpha \in A_n} \langle h, u_\alpha \rangle u_\alpha$. By monotone convergence theorem, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\sum_{\alpha \in A} |\langle h, u_\alpha \rangle|^2 < \sum_{\alpha \in A_N} |\langle h, u_\alpha \rangle|^2 + \varepsilon$$

Thus, for $m \geq n \geq N$

$$\begin{aligned}
\|g_m - g_n\|^2 &= \| \sum_{\alpha \in A_m \setminus A_n} \langle h, u_\alpha \rangle u_\alpha \|^2 \\
&= \sum_{\alpha \in A_m \setminus A_n} |\langle h, u_\alpha \rangle|^2 \\
&= \sum_{\alpha \in A_m} |\langle h, u_\alpha \rangle|^2 - \sum_{\alpha \in A_n} |\langle h, u_\alpha \rangle|^2 \\
&= \sum_{\alpha \in A} |\langle h, u_\alpha \rangle|^2 - \sum_{\alpha \in A_n} |\langle h, u_\alpha \rangle|^2 \\
&< \varepsilon
\end{aligned}$$

Therefore, we conclude that (g_n) is a Cauchy sequence. By completeness of $\mathcal{H}$, $g_n \to g$ for some $g \in \mathcal{H}$. Let $\gamma = h - g$. If $\gamma \in A_o$, then $\gamma \in A_n$ for some $n \in \mathbb{N}$. Thus we'll get that

$$\langle h - g, u_\alpha \rangle = 0$$

verify

If $\gamma \notin A_o$, then $\langle h, u_\gamma \rangle = 0$ and $\langle g_n, u_\gamma \rangle = 0$, so again $\langle , \rangle$ verify

$\square$

16/01/2025

Exercise 1.1.1. Find an orthonormal basis for $\ell^2(X)$

Solution 1.1.1. *We want an orthonormal family $(u_\alpha)_{\alpha\in A}$ such that*

$$\|h\|^2 = \sum_{\alpha\in A} |\langle h, u_\alpha\rangle|^2$$

But here $\|h\|^2 = \sum_{x\in X} |h(x)|^2$, and

$$\langle h, g\rangle = \sum_{x\in X} h(x)\overline{g(x)}$$

If we choose $u_x = \chi_x$, then we see that this satisfy our required properties.

We formulate consequences of the characterization of orthonormal bases.

Theorem 1.1.3. *Let $(u_\alpha)_{\alpha\in A}$ be an orthonormal basis for a Hilbert space $\mathcal{H}$. Let*

$$A_o = \{\alpha \ : \ \langle h, \alpha\rangle \neq 0\}$$

Use the connection between inner product and evaluation from above to see that A_o is at most countable.

Then enumerating A_o by $A_o = \{\alpha_1, \alpha_2, \ldots\}$ and considering

$$h_n = \sum_{j=1}^{n} \langle h, u_{\alpha_j}\rangle u_{\alpha_j}$$

gives $h_n \to h$.

Proof. We saw that h_n forms a cauchy sequence bases on the last oproof and choose

$$A_j = \{\alpha_k \ : \ 1 \leq k \leq j\}$$

Let $g = \lim_{n\to\infty} h_n$. Then the continuity of the inner product gives $\langle h - g, u_\alpha\rangle = 0$ for each $\alpha \in A$. Thus

$$\|h - g\|^2 = \sum_{\alpha\in A} |\langle h - g, u_\alpha\rangle|^2 = 0$$

so $h = g$. □

Corollary 1.1.3.1. *If $(u_\alpha)_{\alpha\in\mathbb{N}}$ is an orthonormal basis for a Hilbert space $\mathcal{H}$, then for each $h \in \mathcal{H}$,*

$$h = \sum_{n\in\mathbb{N}} \langle h, u_n\rangle u_\alpha$$

Conversely, if $\sum_{n\in\mathbb{N}} |c_n|^2 < \infty$, then

$$\sum_{n\in\mathbb{N}} c_n u_n \in \mathcal{H}$$

Proof. verify $\square$

Remark 1.1.1. We can also take $\mathbb{Z}$ instead of $\mathbb{N}$ in the above cases.

Example 1.1.1. Consider $L^2([-\pi, \pi])$, then

$$e_n(x) = \frac{1}{\sqrt{2\pi}} e^{inx}$$

defines an orthonormal basis $(e_n)_{n\in\mathbb{Z}}$ for $L^2([-\pi, \pi])$

Proof. It is clear that the above is an orthonormal family. We show that finite linear combinations of e_ns are dense in $L^2([-\pi, \pi])$. Find a $g \in C([-\pi, \pi])$, $\varepsilon/3$ away from g. Now find a h periodic which is $\frac{\varepsilon}{3}$ away from g. Now use Stone-Weierstrass theorem. $\square$

Corollary 1.1.3.2 (Reisz-Fischer Theorem)**.** *If $f \in L^2([-\pi, \pi])$, then*

$$f = \sum_{n\in\mathbb{Z}} \langle f, e_n\rangle e_n$$

and $\|f\|^2 = \sum_{n\in\mathbb{Z}} |\langle f, e_n\rangle^2|$. Moreover, if $c \in \ell^2(\mathbb{Z})$, then

$$g = \sum_{n\in\mathbb{Z}} c_n e_n \in L^2([-\pi, \pi])$$

Remark 1.1.2. For $f \in L^2([-\pi, \pi])$, $c_n = \langle f, e_n\rangle$ are called Fourier coefficients.

Theorem 1.1.4. *Let $\mathcal{H}$ be a Hilbert space, then $\mathcal{H}$ has an orthonormal basis.*

Proof. We will show this using Zorn's lemma. Let

$$\mathscr{S} = \{U \subset H \ : \ U \text{ is an orthonormal set}\}$$

It is easy to see that $\mathscr{S}$ is nonempty. Order $\mathscr{S}$ by set inclusion. Let $\mathscr{C}$ be a chain in $\mathscr{S}$, then

$$U_{\mathscr{C}} = \bigcup_{C\in\mathscr{C}} C$$

will be an orthonormal set in $\mathscr{S}$. (Use the standard arguments to see this). Therefore $U_{\mathscr{C}}$ is the upper bound for the chain $\mathscr{C}$. Hence we see that $\mathscr{S}$ has a maximal element by the Zorn's lemma. Hence $\mathcal{H}$ has an orthonormal basis. $\square$

Definition 1.1.4. Let $\mathcal{H}, \mathcal{K}$ be Hilbert space. A linear map $V : \mathcal{H} \to \mathcal{K}$ is an isometry if

$$\langle Vh, Vg \rangle_{\mathcal{K}} = \langle h, g \rangle_{\mathcal{H}}$$

for each $h, g \in \mathcal{H}$. If V is onto, we say V is unitary. Then we say $\mathcal{H}$ and $\mathcal{K}$ are isomorphic. (From preservation of preservation of norm, V is one-one).

Proposition 1.1.1. *A linear map $V : \mathcal{H} \to \mathcal{K}$ is an isometry if and only if for every $h \in \mathcal{H}$,*

$$\|Vh\|_{\mathcal{K}} = \|h\|_{\mathcal{H}}$$

Proof. One way is easy. That is if the inner product is preserved, then the norm is preserved. Conversely, we use the polarization identity.

$$\langle h, g \rangle = \frac{1}{4} \sum_{j=1}^{4} i^j \|h + i^j g\|^2, \quad (i = \sqrt{-1})$$

□

Example 1.1.2. Let $\mathcal{H} = \ell^2(\mathbb{N})$. Define

$$S : \mathcal{H} \to \mathcal{H} := (x_1, x_2, \ldots) \mapsto (0, x_1, x_2, \ldots, x_n)$$

Then S is clearly an isometry, but not unitary since nothing maps to $(1, x_1, x_2, \ldots)$

23/01/2025

Example 1.1.3. Given an infinite dimensional Hilbert space with orthonormal basis $(u_n)_{n \in \mathbb{N}}$, show that $\{u_n\}$ is not compact.

Proof. Since $\|u_\alpha - u_\beta\| = \sqrt{2}$, take $\frac{1}{\sqrt{2}}$ radius balls around each u_α to get a collection of open balls that cover the set with no finite subcover.

Another way to see is to use the sequential compactness criterion and see that the sequence (u_n) does not have any convergent subsequence. Since this is a metric space, sequential compactness is equivalent to compactness. □

Theorem 1.1.5 (Every Hilbert space is $\ell^2(A)$)**.** *Let $\mathcal{H}$ be a Hilbert space, $(u_\alpha)_{\alpha \in A}$ is an orthonormal basis, then there is a unitary map $U : \mathcal{H} \to \ell^2(A)$ such that $U(u_\alpha) = \chi_\alpha$*

Proof. We first note that by linearity, if $p \in \operatorname{span}\{u_\alpha \ : \ \alpha \in A\}$, then $U(p)$ is determined by χ_α. Next, by Bessel's inequality,

$$\|U(p)\|_{\ell^2(A)} \leq \|p\|$$

Hence U is bounded. Hence it can be continuously extended to $\mathcal{H} = \overline{\operatorname{span}\{u_\alpha\}}$ as a limit of sequences. Also, by the equivality in the Bessel's inequality, we get that U is an isometry, hence one-to-one.

Now it remains to show that U is onto. Given $g \in \ell^2(A)$, we know that there exists at most a countable set $\{\alpha_1, \alpha_2, \ldots\} = A_0$ such that $g(\alpha_i) \neq 0$. Consider

$$h_n = \sum_{j=1}^{n} g(\alpha_j) u_{\alpha_j}$$

then,

$$u(h_n)(\alpha) = \begin{cases} g(\alpha_j), & \alpha_j \in A_0 \\ 0, & \text{otherwise} \end{cases}$$

Moreover,

$$\left\| U(h_n) - g \right\|_{\ell^2(A)}^2 = \sum_{j=n+1}^{\infty} |g(\alpha_j)|^2 \to 0$$

Now if

$$h = \sum_{j=1}^{\infty} g(\alpha_j) u_{\alpha_j} \in \mathcal{H} \quad (\text{ since } g \in \ell^2(A))$$

we get

$$\|h - h_n\|^2 = \|U(h_n) - g\|_{\ell^2(A)}^2 \to 0$$

and the injectivity of U shows that $U(h) = g$. $\square$

Chapter 2

Banach Space Techniques

Definition 2.0.1. If X is a real or complex normed vector space with a norm, and the complete in the topology induced by the norm, it is called a Banach space.

Definition 2.0.2. If X, Y are normed vector spaces over $\mathbb{R}$ or $\mathbb{C}$, $\Lambda : X \to Y$ linear, then the norm of the operator

$$\|\Lambda\| = \sup\{\|\Lambda x\| \ : \ \|x\| < 1\}$$

If $\|\Lambda\| < \infty$, then we say that Λ is bounded.

Proposition 2.0.1. *Given $\Lambda : X \to Y$, a linear map between normed linear spaces, the following are equivalent*

(1) Λ is bounded

(2) Λ is continuous

(3) Λ is continuous at some $x_o \in X$

Proof. $(1 \implies 2)$

$$\|\Lambda(x - y)\| \leq \|\Lambda\|\|x - y\|$$

gives $\|\Lambda\|$-Lipschitz continuity.

$(2 \implies 3)$ Follows from the definition.

$(3 \implies 1)$ For each $\varepsilon > 0$, there is $\delta > 0$ such that for each $x \in X$, with $\|x - x_o\| < \delta$, then $\|\Lambda x - \Lambda x_o\| < \varepsilon$. Thus for $\|y\| < \delta$, by linearity of Λ, we get

$$\|\Lambda y\| = \|\Lambda(x_o + y) - \Lambda x_o\| < \varepsilon$$

Again using linearity, we get for $\|y'\| < 1$,

$$\|\Lambda y'\| < \frac{\varepsilon}{\delta} < \infty$$

Now since $\overline{B_1(0)} \subset B_2(0)$, we see that $\|\Lambda\| < \frac{2\varepsilon}{\delta} < \infty$. $\square$

Theorem 2.0.1 (Baire Category Theorem)**.** *If (X, d) is a complete metric space, and $V_1, V_2, \ldots$ are dense subsets, then*

$$\bigcap_{n=1}^{\infty} V_n$$

is dense in X.

Proof. We show that for any non-empty open set $W \subset X$,

$$\bigcap_{n=1}^{\infty} V_j \cap W \neq \emptyset$$

We write $B_r(x) = \{y \in X \ : \ d(x, y) < r\}$. Since V_1 is dense and open, $V_1 \cap W$ is open and dense in W. Thus we can find an $r_1 > 0$ such that $\overline{B_{r_1}(x_1)} \subset W \cap V_1$. (First find an $r' > 0$ such that $B_{r'}(x_1) \subset W \cap V_1$. Then take $r_1 = \frac{r'}{2}$).

We inductively proceed by taking $x_n \in V_n \cap B_{r_{n-1}}(x_{n-1})$ such that $\overline{B_{r_n}(x_n)} \subset V_n \cap B_{r_{n-1}}(x_{n-1})$. Without loss of generality, choose $0 < r_n < \frac{1}{n}$. This gives a sequence which satisfies for $i, j > n$ that $x_i, x_j \in B_{r_n}(x_n) \implies d(x_i, x_j) < 2r_n < \frac{2}{n}$. Hence x_n is Cauchy. By completeness $x_n \to x \in \overline{B_{r_n}(x_n)} \subset V_n \cap W$ for all n. Thus $x \in W$ and

$$x \in \bigcap_{n=1}^{\infty} V_n$$

□

28/01/2025

Corollary 2.0.1.1. *Let $G_1, G_2, \ldots$ be a sequence of dense G_δ subsets, then $\cap_{n=1}^{\infty} G_n$ is dense and G_δ.*

Theorem 2.0.2 (Banach-Steinhaus theorem)**.** *If X is a Banach space and Y a normed vector space. Let $(T_\alpha)_{\alpha \in A}$ be a family of bounded linear maps from $X \to Y$. Then either of the two holds,*

(1) $\exists M \geq 0$ such that $\|T_\alpha\| \leq M$ for all $\alpha \in A$.

(2) The set $\{x \in X \ : \ \sup_\alpha \|T_\alpha x\| = \infty\}$ is a dense G_δ set.

Corollary 2.0.2.1. *With X, Y, T_α as above, if for each $x \in X$, $\sup_{\alpha \in A} \|T_\alpha x\| < \infty$, then there is a $M \geq 0$ such that*

$$\sup_{\alpha \in A} \|T_\alpha\| \leq M$$

We study consequences before looking at the proof of Theorem 2.0.2.

Exercise 2.0.1. Suppose (a_n) is a sequence such that for each $(b_n) \in \ell^2(\mathbb{N})$, $\sum_{n\in\mathbb{N}} a_n b_n < \infty$, then $a \in \ell^2(\mathbb{N})$.

Proof. To see this, take

$$T_n : \ell^2(\mathbb{N}) \to \mathbb{C} := T_n(b) \mapsto \sum_{i=1}^{n} \overline{a_i} b_i$$

and observe

$$\begin{aligned} |T_n(b)| &\leq \Big(\sum_{i=1}^{n} |a_i|^2\Big)^{\frac{1}{2}} \Big(\sum_{i=1}^{n} |b_j|^2\Big)^{\frac{1}{2}} \\ &\leq \Big(\sum_{i=1}^{n} |a_i|^2\Big)^{\frac{1}{2}} \|b\| \end{aligned}$$

Hence each T_n is linear (by definition) and bounded by the inequality above. Let $(b_n) \in \ell^2(\mathbb{N})$, then for $n \in \mathbb{N}$, we get

$$\begin{aligned} |T_n(b)| &\leq \sum_{i=1}^{n} |a_i b_i| \\ &\leq \sum_{i=1}^{\infty} |a_i b_i| < \infty \end{aligned}$$

Now a direct application of Theorem 2.0.2 gives that three is a $M \geq 0$ such that $|T_n b| \leq M\|b\|$ for each $n \in \mathbb{N}$. Hence if we consider

$$T : \ell^2(\mathbb{N}) \to \mathbb{C} := b \mapsto \sum_{i=1}^{\infty} a_i b_i$$

we find that $\|T\| \leq M$. By Reisz representation theorem for Hilbert space s, there is a $c \in \ell^2(\mathbb{N})$ such that $T(b) = \langle b, c\rangle$. Choosing $b = e_m$, we get $T(e_m) = \bar{c_m} = a_m$. We conclude that $a \in \ell^2(\mathbb{N})$. □

Proof of Banach-Steinhaus Theorem 2.0.2. Let $\phi_\alpha : X \to [0, \infty) := x \to \|T_\alpha x\|$. Look at

$$\begin{aligned} |\phi_\alpha(x) - \phi_\alpha(y)| &= |\|T_\alpha x\| - \|T_\alpha y\||\\ &\leq \|T_\alpha(x-y)\| \\ &\leq \|T_\alpha\|\|x-y\| \end{aligned}$$

Hence ϕ_α is (lower-semi) continuous.

Therefore, we can define a lower semi-continuous function

$$\phi(x) = \sup_{\alpha \in A} \phi_\alpha(x)$$

Thus, for $n \in \mathbb{N}$, $V_n = \{x \in X \ : \ \phi(x) > n\}$ is open. If each V_n is dense in X, then $G = \cap_{n=1}^{\infty} V_n$ is a dense G_δ set, and $\phi(G) = \{\infty\}$. Otherwise, if for some $n \in \mathbb{N}$, one of V_n is not dense. Then that particular V_n^c contains a non-empty open set. Choosing a $B_\delta(y) \subset W$ centered at y, we get that for $x \in X$, $\|x - y\| < \delta$, we have $\phi_\alpha(x) = \|T_\alpha x\| \leq n$ for each $\alpha \in A$. This implies that there is an $M \geq 0$ for which

$$\sup_{\alpha \in A} \|T_\alpha\| \leq M$$

□

We investigate more consequences of Baire's theorem.

Theorem 2.0.3 (Open mapping theorem)**.** *Let X, Y be Banach spaces, $T : X \to Y$ is bounded, linear. If T is onto and $U \subset X$ is open, then $T(U)$ is open.*

Proof. We claim it is equivalent to show $T(B_1(0)) \subset B_\delta(0)$ for some $\delta > 0$. The statement then follows by choosing for U open, a vector $u \in U$ with $\varepsilon > 0$ such that $B_\varepsilon(u) \subset U$. In that case, if $y \in Y$ satisfies

$$\|y - Tu\| < \varepsilon\delta$$

or

$$\|\frac{y - Tu}{\varepsilon}\| < \delta$$

By the inclusion, we get $z \in X$ such that $\|z\| < 1$ and $Tz = \frac{y - Tu}{z}$. Solving for y, gives $y = T(\varepsilon z + y)$. Letting $w = \varepsilon z + u$, then $\|w - u\| = \varepsilon\|z\| < \varepsilon$ and $Tw = y$. We have found for each y near Tu a vector $w \in U$ which maps to y.

Let $U \subset X$ be open. Fix $u \in U$. Then there is $\delta > 0$ with $\{x \ : \ \|x - u\| < \delta\} \subset U$. We also know that if $y \in Y$ □

Corollary 2.0.3.1. *If T is one-one and onto, then T^{-1} is bounded.*

Proof. □